

DEPROPANIZER COLUMN VER 1

Operator Training Simulator · C-201

Chulalongkorn University · Department of Chemical Engineering

DOCUMENT NO. : DC3-V1-IN-CE-001

DOCUMENT TITLE : CAUSE AND EFFECT DIAGRAM

Teaching document generated from the simulator logic (bmsTrip, tripConditions, alarm list).

Rev.	Date	Revision status	Prepared	Checked	Approved	Remark
R0	03-Oct-26	Issued for teaching	A. Soottitantawat	—	—	

DEPROPANIZER COLUMN VER 1 · C-201 CAUSE and EFFECT DIAGRAM Chulalongkorn University · Department of Chemical Engineering								Client :		Chulalongkorn University · Department of Chemical Engineering						REV	DATE	ORIGINATOR	CHECKER	APPROVER	
								PROJECT :		Depropanizer Column Ver 1 · OTS						R0	03-Oct-26	A. Soottitantawat	—	—	
								DOCUMENT No. :		DC3-V1-IN-CE-001											
								DOCUMENT TITLE :		Cause and Effect Diagram											
								P&ID :		P&ID-001 rev C / DC3-V1-PID-001 rev D											SHEET 1 of 1
Causes								EFFECTS													
								XV-201 · Feed shut-off valve (SIS)	FIC-201 · Feed flow controller → MAN 0 % (FV-201 closes)	XV-202 · Preheater E-201 steam shut-off (SIS)	XV-209 · Reboiler E-203 steam shut-off (SIS)	TIC-202 · Preheat controller → MAN 0 % (TV-202 closes)	TIC-208 · Tray 8 controller → MAN 0 % (TV-208 closes)	P-201B · Spare feed pump (turbine) auto-start	P-202B · Spare reflux pump auto-start	SIS · SIS state	1ST · First-out annunciation	AMB · Flashing lamp (amber)	RED · Flashing lamp (red)		
Trip Action								C	MAN	C	C	MAN	MAN	ST	ST	TP/RS	F	A	A	A	
Item No.	Tag No.	Tag Service Description	Trip/Alarm-Direction	Setting	Eng. Unit	Delay	Note no.													Notes:	
001	HS-ESD	Emergency shutdown push-button (DCS / SIS screen)	Tripped (opened contact)	—	—	0 s	4	X	X	X	X	X	X			TP	X		X	1. Action codes: C = close; MAN = controller forced to manual with output 0 %; ST = start; TP = SIS tripped (manual reset required); RS = SIS stays in IDLE; F = first-out; A = activate. Brackets = recommended row, not in the logic. 2. LSLL-202 is armed only while heating is enabled (SIS in RUN); LSHH-203 and the ESD are always armed. 3. Trip effects latch until reset. Reset only when no LSHH-203 cause remains; the SIS then returns to IDLE with XV-201 open, and the operator re-opens XV-209 and XV-202 with the start permissives (note 10). 4. Manual trip; immediate, no delay. Same effects as LSHH-203 (feed and heat off). 5. Alarm suppressed while heating is off (SIS not in RUN), as in the simulator alarm list (suppression by state). 6. Alarm suppressed with no feed or with heating off. 7. Rev B upgrade only (FAL-201, LAL/LAH-201); not active on the design case rev A. 8. Auto-starts, not trips: FSLL-201 acts only with P-201B stopped, XV-201 open, FV-201 above 20 % and a feed pump running or just tripped; PDSL-230 only with P-202B stopped and a reflux pump running or just tripped. Each acts once, then disarms until re-armed on the field screen. Nothing is shut down. 9. PSHH-221 is the recommendation of the HAZOP, Safety layers and LOPA tabs (setting 1.77 MPa g assumed): it would cut the heat like LSLL-202. PSV-201 (set 1.80 MPa g) is a mechanical device and is not on this sheet. 10. Start permissives: SIS reset (IDLE), cooling water FI-216 ≥ 250 m³/h, sump level LIC-202 ≥ 30 %, TIC-208 and TIC-202 in MAN with output ≤ 10 %, steam block valves HV-209 and HV-202 open. Not a trip: no valve moves. 11. Not tripped by the SIS: feed pumps P-201A/B, reflux pumps P-202A/B, cooling water, PIC-210, LIC-202, LIC-203 and FIC-204 stay as they are after a trip.	
002	FIC-201	Feed flow (rev B FAL-201) low alarm	Low	45	m³/h		7											X	X		
003	FIC-201	Feed flow (rev B FAL-201) high alarm	High	70	m³/h		7											X	X		
004	FSLL-201	Feed flow low-low (FT-201): spare feed pump P-201B auto-start	Low-Low	15	m³/h	5 s	8							X				X	X		
005	TIC-202	Feed preheat low alarm	Low	68	°C		6											X	X		
006	TIC-202	Feed preheat high alarm	High	85	°C		6											X	X		
007	PI-207	LP steam header low alarm	Low	350	kPa g													X	X		
008	PIC-210	Column pressure low alarm	Low	1.60	MPa g		5											X	X		
009	PIC-210	Column pressure high alarm	High	1.76	MPa g		5											X	X		
010	PI-211	Column pressure high (independent) alarm	High	1.78	MPa g													X	X		
011	PSHH-221	Column pressure high-high trip on its own PT-221 (RECOMMENDED, not in the logic)	High-High	1.77	MPa g	3 s	9			(X)	(X)	(X)	(X)			(TP)	(X)		(X)		(X)
012	TIC-208	Tray 8 temperature low alarm	Low	72	°C		5											X	X		
013	TIC-208	Tray 8 temperature high alarm	High	82	°C		5											X	X		
014	TIC-227	Top tray temperature high alarm	High	50	°C		5											X	X		
015	PDI-212	Column ΔP high (flooding) alarm	High	30	kPa													X	X		
016	LIC-202	C-201 sump level low alarm	Low	25	%													X	X		
017	LIC-202	C-201 sump level high alarm	High	75	%													X	X		
018	LSLL-202	C-201 sump level low-low trip (LS-202): heat off	Low-Low	5	%	5 s	2, 3			X	X	X	X			TP	X		X		X
019	LIC-203	Reflux drum level low alarm	Low	25	%													X	X		
020	LIC-203	Reflux drum level high alarm	High	75	%													X	X		
021	LSHH-203	Reflux drum V-202 level high-high trip (LS-203): feed and heat off	High-High	92	%	5 s	3	X	X	X	X	X	X			TP	X		X		X
022	FIC-204	Reflux flow low alarm	Low	35	m³/h		5											X	X		
023	FIC-204	Reflux flow high alarm	High	65	m³/h		5											X	X		
024	PDSL-230	Reflux pump differential pressure low (PDT-230): spare reflux pump P-202B auto-start	Low	150	kPa	3 s	8								X			X	X		
025	AI-219	Propane product off spec (C4+) high alarm	High	2.50	mol %		6											X	X		
026	AI-220	Bottoms off spec (C3) high alarm	High	2.00	mol %		6											X	X		
027	FI-216	Cooling water flow low alarm	Low	250	m³/h		5											X	X		
028	LI-201	Feed drum V-201 level (rev B LAL/LAH-201) low alarm	Low	20	%		7											X	X		
029	LI-201	Feed drum V-201 level (rev B LAL/LAH-201) high alarm	High	80	%		7											X	X		
030	I-201	Open XV-209 / XV-202 refused: start permissive not met	Not met	—	—		10									RS		X			
031																					
032																					

Notes:

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